Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?Answer Choices:A. It activates TLR5 on the surface of macrophages.B. It primarily induces the Th1-cell response.C. Peptidoglycan is its major antigen that induces an immune response.D. This pathogen evades the immune response by encapsulation.

Reasoning + Answer:

<think>The patient presents with symptoms of community-acquired pneumonia, including a sore throat, cough, fever, and radiographic evidence of patchy reticular opacities in the lungs. The patient was diagnosed with community-acquired pneumonia and treated with cephalexin, which is an antibiotic. This indicates that the pathogen is likely a bacterial species, such as Streptococcus pneumoniae. The immune response to bacterial pathogens typically involves the activation of TLRs (Toll-like receptors) and Th1-cell responses. Peptidoglycan is a major component of the bacterial cell wall, which is recognized by TLR5 on the surface of macrophages. The pathogen Streptococcus pneumoniae does not encapsulate and evade the immune response; rather, it is recognized by the immune system through its peptidoglycan and other bacterial components. A. It activates TLR5 on the surface of macrophages - This is a correct response as peptidoglycan in bacteria is recognized by TLR5. B. It primarily induces the Th1-cell response - This is also correct, as bacterial infections often elicit a Th1-type immune response. C. Peptidoglycan is its major antigen that induces an immune response - This is correct as peptidoglycan is a major component of bacterial cell walls and is recognized by the immune system. D. This pathogen evades the immune response by encapsulation - This is incorrect, as Streptococcus pneumoniae does not encapsulate to evade the immune response. Considering the context of the patient's symptoms, diagnosis, and treatment, the most direct and relevant immune response activated by the bacterial pathogen is the recognition of its peptidoglycan component. </think><answer>C. Peptidoglycan is its major antigen that induces an immune response. </answer>